

Name Uniqueness and the rise of individualism in the Western hemisphere (1500 - 2000)

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ABSTRACT

This study investigates the historical evolution of individualism by analysing trends in Name Uniqueness — the distinctiveness of personal names — as a proxy for values emphasising autonomy, independence and self-expression. Using a dataset of 2313,713 notable individuals from Wikidata (1500–2000), we aimed to capture the historic evolution of Name Uniqueness in the Western hemisphere. Our analysis reveals an increase in individualism in Europe, the Americas, and Australia that predates widespread modernisation, which aligns with the conclusions of historians. In line with modernisation theory, we also observe a strong positive correlation between GDP per capita and Name Uniqueness across five centuries. These results remain robust when controlling for gender, occupation, and time period. Our findings challenge Max Weber's thesis that the spread of Protestantism is associated with an increase in individualism; in fact, we find that it is associated with a decrease. By combining large-scale historical data with a validated cultural indicator, this study contributes to the growing body of evidence linking economic development to profound shifts in identity and social behaviour. It highlights Name Uniqueness as a powerful tool for tracing the long-term dynamics of individualism in large scale societies.

1. Introduction

Individualism is characterised by a focus on personal autonomy, independence and self-expression. In such societies, people tend to define themselves primarily by their personal achievements, goals, and unique attributes. This perspective contrasts with that of collectivist cultures, where personal aspirations are often subordinated to maintaining group harmony and adhering to societal expectations (Hofstede, 1984; Markus and Kitayama, 1991; Nisbett et al., 2001; Peng and Nisbett, 1999).

Over the past 50 years, large-scale international surveys have revealed a global increase in individualism, which is closely tied to the broader process of modernisation (Bianchi, 2016; Inglehart, 1990). In high-income Western countries such as the United States and much of Europe, individualism has become more prevalent, characterised by a stronger focus on personal autonomy, self-expression and independence (Inglehart, 2018; Santos et al., 2017). This shift towards individualism is also evident, albeit to a lesser extent, in non-Western nations such as China (Bao et al., 2024; Cai et al., 2018) and Japan (Ogihara, 2021; Ogihara et al., 2015), where traditional collectivist values are gradually yielding to individualistic behaviours (Ogihara, 2017).

The global rise of individualism can also be observed in other ways. For example, studies have shown an increase in the use of the first-person singular pronoun “I” and a decline in the use of the plural pronoun “we” in American English books from 1960 to 2008 (Twenge et al., 2013). Similarly, stylistic choices such as clothing and hairstyles also highlight this trend. For example, an analysis of over 14 million US high school yearbook photos from 1950 to 2020 revealed a decline in conformity and an increase in individualistic expression, particularly after the 1960s (Yanagizawa-Drott and Hans-Joachim, 2024). Furthermore, research into naming practices has revealed a preference for unique and unconventional names in various countries, including Japan (Ogihara, 2021), China (Bao et al., 2021; Ogihara, 2023), the US (Lieberman and Bell, 1992; Twenge et al., 2010, 2016), Germany (Gerhards and Hackenbroch, 2000) France, UK and Brazil (Mignot, 2022). Together, these findings provide evidence of an increasing emphasis on self-expression and personal autonomy throughout the 20th century.

Due to limited data, most analyses focused on the contemporary era. But how did individualism manifest before the 20th century? Many non-industrial, small-scale societies exhibit high levels of personal autonomy (Jackson et al., 2020). By contrast, historians, since Jacob Burckhardt (1860) and Max Weber (1930), have frequently noted that large-scale

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premodern societies in Europe, North Africa and Asia were generally characterized by limited personal autonomy, minimal self-expression, and robust norms of obedience and conformity. Individualism began to rise significantly during the late medieval and early modern periods, a trend reflected across various domains (Gurevich, 1995; MacFarlane, 1988; McCloskey, 2007, 2017; Schulz et al., 2019). In the arts, the Renaissance marked a shift toward individualism, with artists such as Leonardo da Vinci and Michelangelo emphasizing personal authorship by signing their works. Portraiture, too, began to focus on individual identity and character, moving away from purely religious or collective themes (Jong-crane and Liebrechts, 1998; Martin, 1997). The Protestant Reformation further propelled this shift, as figures like Martin Luther underscored personal faith and a direct relationship with the divine, encouraging individuals to interpret religious texts independently and thereby fostering spiritual autonomy (McCloskey, 2007, 2017; Weber, 1930). In Western literature, the focus shifted from collective ideals to individual emotions and desires as central narrative drivers. Sentimental novels such as Samuel Richardson's *Pamela* (1740) and Jean-Jacques Rousseau's *Julie, or the New Heloise* (1761) explored characters' inner lives and emotional experiences with unprecedented depth. The Romantic movement of the late 18th and early 19th centuries further amplified these themes, celebrating individual autonomy, personal freedom, and the defiance of social conventions in pursuit of self-fulfillment. This is exemplified in Goethe's *The Sorrows of Young Werther* (1774) (Izenberg, 1992; Wulf, 2022).

Despite the extensive scholarship on the historical rise of individualism, a significant gap remains: the lack of quantitative estimates to track its evolution over time. While qualitative research provides detailed narratives and insightful case studies, it often fails to answer critical questions with precision. When did individualism begin to rise? Was its growth steady and linear, or did it occur in bursts influenced by historical events? Did this increase happen uniformly across Europe, or were there significant regional differences? Quantitative analysis is crucial for answering these questions and deepening our understanding of the origins of individualism. One long-standing hypothesis, proposed by Max Weber (1930), suggests that Protestantism played a pivotal role in fostering modern values, particularly individualism and personal autonomy. He argued that Protestantism's individualistic ethos encouraged people to interpret Christianity for themselves, laying the foundations for greater individual freedom and social change. A quantitative approach could determine whether the adoption of Protestantism in the 16th century coincided with a measurable increase in individualism and whether the rise of individualism is specific to Protestant countries.

Furthermore, sociological research originally conducted by Ronald Inglehart (1990, 2018) has repeatedly shown a significant correlation between individualism and economic development. This body of work shows that societies tend to shift towards individualistic values, emphasising personal freedom and self-expression over survival needs, at high levels of economic development (Welzel, 2013). However, it is unclear whether this association held true before 1950. By quantifying individualism over time and across regions, we can explore these connections and identify the factors that drove the emergence of individualistic values.

Personal names provide a continuous and remarkably detailed record deep into history (de la Croix and Licandro, 2015; Laouenan et al., 2022). Moreover, as we have seen, names can serve as a tool for recovering psychological traits and individual values. In particular, "Name Uniqueness" has been extensively studied in social psychology and is widely recognized as a reliable indicator of individualism (Baumard et al., 2024; Varnum and Kitayama, 2011). A recent study by Knudsen (2019) further solidifies this connection, demonstrating that unique names are strongly correlated with individualistic values. Analyzing a dataset of baby names from 2015, spanning 38 countries and 13 subnational regions, Knudsen found a positive relationship between the prevalence of unique first names and individualistic cultural

traits. Specifically, countries with a higher proportion of unique names tend to exhibit greater individualism, even after controlling for factors such as ethnic diversity and other cultural dimensions. Moreover, within-country analyses using data from the World Values Survey and the European Values Study reveal that regions with more unique first names show stronger preferences for individualistic traits like autonomy and self-expression, as opposed to collectivistic traits such as obedience and conformity. These findings reinforce the validity of Name Uniqueness as a robust measure of individualism.

In this paper, we examined the first names recorded on the Wikidata platform between 1500 and 2000. Wikidata is a rich and reliable source for studying historical figures. The database integrates data from multiple language editions of Wikipedia and Wikidata, offering coverage of over two million unique individuals. This approach captures an elite subset of historical figures, representing around one in 43,000 people who have ever lived. Furthermore, the data collected from both sources undergoes cross-verification and deduplication to ensure a high level of accuracy, particularly for well-documented individuals. While sparse information and classification ambiguity present challenges for lesser-known figures, the database employs rigorous error-detection methods. Manual verification and algorithmic corrections reduce errors, making the dataset highly reliable for statistical and historical analyses. These features make the database an invaluable resource for social scientists, historians and cultural researchers seeking to study the historical impact of notable individuals using quantitative methods (Laouenan et al., 2022).

2. Material and methods

In practice, this study examines the association between Name Uniqueness as a proxy for Individualism and GDP per capita as a proxy for economic growth. To assess historical Name Uniqueness scores, we collected all instances of individuals with a valid name, year of birth, and place of birth from Wikidata. After data cleaning, we obtained 2313,713 historical individuals born from 1500 to 2000 (Fan et al., 2024). For economic growth, we used the latest historic GDP estimate, providing 21,703 observations of annual GDP per capita for countries and regions (Koch et al., 2024).

2.1. Calculating name uniqueness

To use names as a proxy for individualism, we created a single continuous index that captures the relative rareness of names within a given community. This "Name Uniqueness Score" is measured by calculating the mean value of name frequencies over a specified time period (Bao et al., 2021, 2024). More precisely, the value is computed by dividing the frequency of a specific name in a given time period by the total number of names recorded during that period. For example, if the name "Jean" was given to 10 infants out of a total of 100 unique names in a 10-year period, the frequency of the name "Jean" for that decade would be 0.1. This frequency is then log-normalized to produce the "Name Uniqueness Score."

To calculate the Name Uniqueness Score, we follow these steps:

1. Let n_i be the number of instances of a name i within a specified time period.
2. Let N be the total number of names recorded during the same time period.
3. Calculate the frequency of name i as $\frac{n_i}{N}$.
4. We add a very small number, 10^{-6} , to ensure that the argument of the logarithm is always positive and that the uniqueness scores for names would not be heavily skewed towards too-infrequent names.
5. Apply log-normalization to the frequency to create a normal distribution and transform the fraction to a positive number. The log-normalized frequency is $\log\left(\frac{n_i}{N} + 10^{-6}\right)$.

6. Take the mean of all individual uniqueness scores within the designated period to produce the Name Uniqueness Score, which serves as an index for individualism of that period.

The Name Uniqueness Score is thus given by:

$$U = \frac{1}{N} \sum_{i=1}^n \left(-\log_{10} \left(\frac{n_i}{N} + 10^{-6} \right) \right)$$

This score provides a quantitative measure of the relative rareness of names within a given time period, which can be used to infer trends in individualism.

2.2. The bootstrapping method

One issue with Name Uniqueness is that the effect size increases as the sample size grows. Intuitively, Name Uniqueness can be represented as a fraction: the numerator is the number of times a name occurs, and the denominator is the total number of names in the dataset. Therefore, the larger the denominator, the smaller the fraction. After log normalisation, this translates into a higher uniqueness score. To demonstrate this, we computed Name Uniqueness for the same population with different sample sizes. As can be seen from the grey dots in Fig. 1, Name Uniqueness increases as sample size grows, which would not be expected if Name Uniqueness were unaffected by sample size.

To address this issue, we employed a bootstrapping method to generate a balanced sample for calculating Name Uniqueness. Firstly, to account for variations in the number of observations across years, we increased the time period used for this calculation from one year to ten years. This increases the sample size and achieves a more consistent distribution. Next, we applied a bootstrapping technique to mitigate the bias introduced by the calculation (see Fig. 2). A particular time period is picked from the database (shown by the red bar in Fig. 2A) and 200 individuals are randomly sampled. We then calculated the uniqueness of

these 200 individuals, resulting in 200 uniqueness values. These values are averaged to obtain a Name Uniqueness score for the time period. This process was repeated 1000 times within the same time period to ensure that all individuals within the period were covered (bootstrap). This results in 1000 uniqueness scores for a single time period. These are then averaged to derive a single aggregate value, which is used as the uniqueness score for the time period (shown by the red dot in Fig. 2B). This approach limits the sample size of the Name Uniqueness calculations to a fixed amount, thus avoiding bias resulting from the effect of an increasing sample size on an increasing effect, while also covering all available data.

This bootstrapping process was then applied to all time periods, generating the longitudinal data used in this study (as shown in Fig. 2B). We also apply a minimum threshold to the calculation, set at 200 individuals. Time periods with fewer than 200 individuals are excluded, as the calculation becomes unreliable with such small sample sizes. We conducted sensitivity analysis using a threshold of 500 and 1000 individuals. The general trend of the data remains similar to the version using the 200 (Fig. S1) thresholds, but the 1000 thresholds significantly reduce the number of available data points. Hence we pick the 200 threshold which produced the maximum number of datapoints.

We also created a confidence interval to estimate the accuracy of our estimate. First, we calculated the standard deviation of the Name Uniqueness score for time periods with over 10,000 individuals. The large sample size ensures a reliable estimation of the standard deviation. We then took the mean of these standard deviations to estimate the standard deviation for the entire dataset ($\sigma = 0.03176$). Using this value, we estimated the 95 % confidence interval for each data point.

$$CI = U \pm 1.96 \times \frac{\sigma}{\sqrt{\frac{n}{200}}}$$

This extended calculation method ensures that the Name Uniqueness measurement is based on a uniform dataset, thereby minimising potential bias caused by fluctuations in the number of observations. The

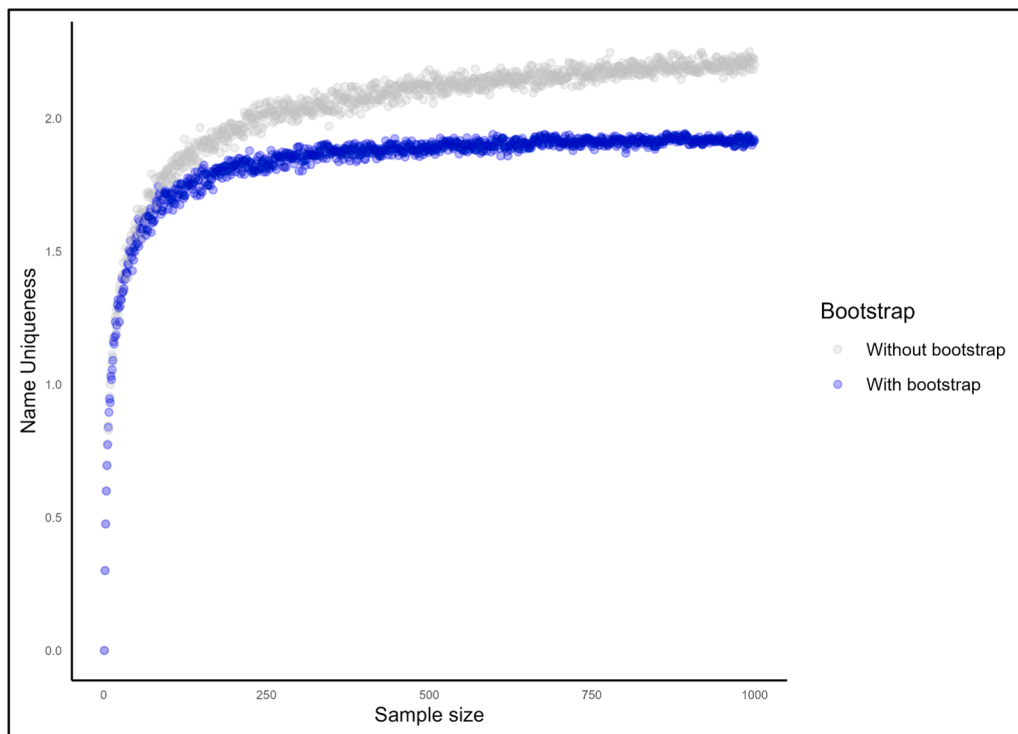


Fig. 1. The relationship between Name Uniqueness (NU) and sample size. Without correction (grey dots), NU increases with sample size, introducing bias. The bootstrap method (blue dots) corrects for this, keeping NU stable across sizes. The population is all names within a 10-year window. The sharp rise in NU below ~200 reflects skewed name distributions: common names dominate small samples, underestimating uniqueness. As sample size grows, rare names are more likely to appear, and NU stabilizes. Around 200–300, the sample begins to reflect the true population distribution.

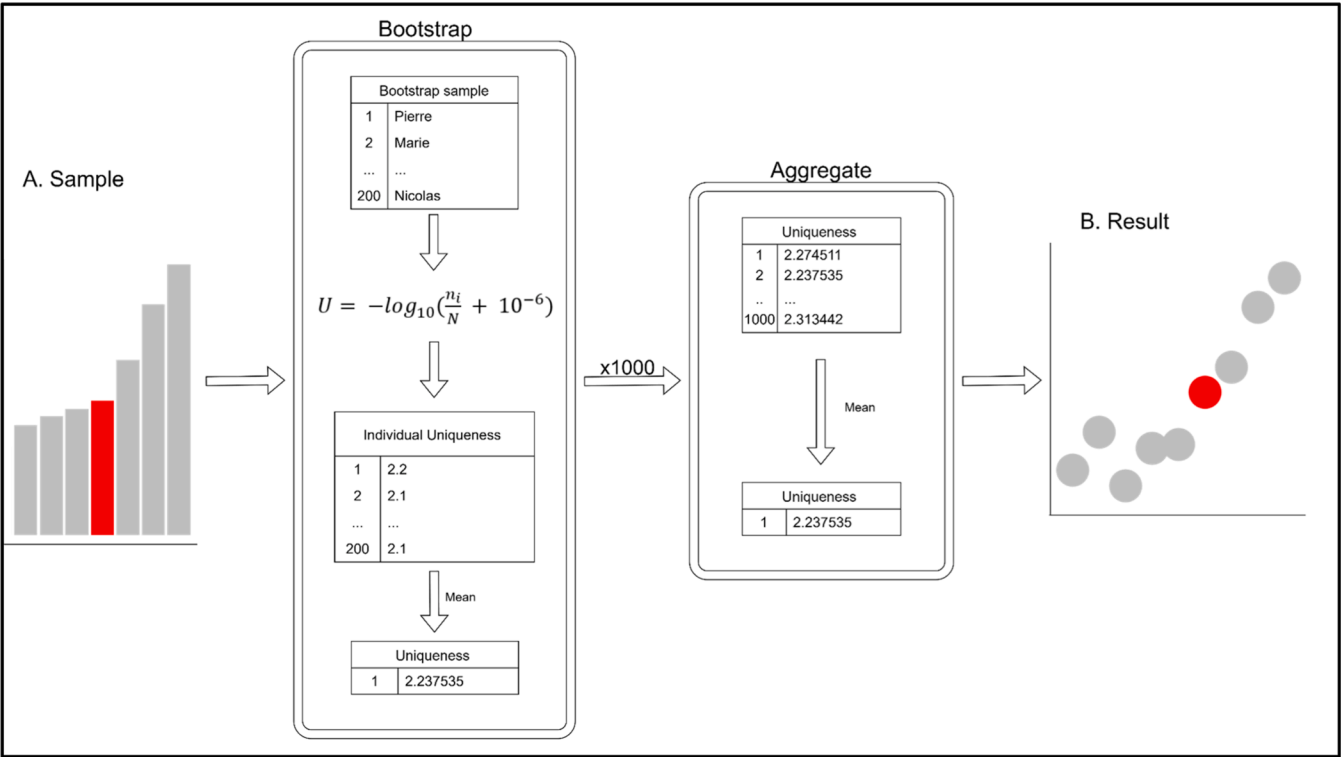


Fig. 2. Bootstrapping process for Name Uniqueness, the process starts with a sample from a specific year (Fig. 2A), it then goes through a process which resamples names every time period for 1000 iterations to produce a single value for that period, resulting in the one datapoint in Fig. 2B. This method controls the bias introduced by fluctuations in sample size across time.

bootstrapping approach stabilises the effect size, ensuring that it remains consistent regardless of the sample size (see the blue dots in Fig. 1). This makes it possible to analyse Name Uniqueness meaningfully

even with significantly smaller sample sizes. Consequently, this method is a valuable tool for studying cultural history, where data limitations often pose significant challenges.

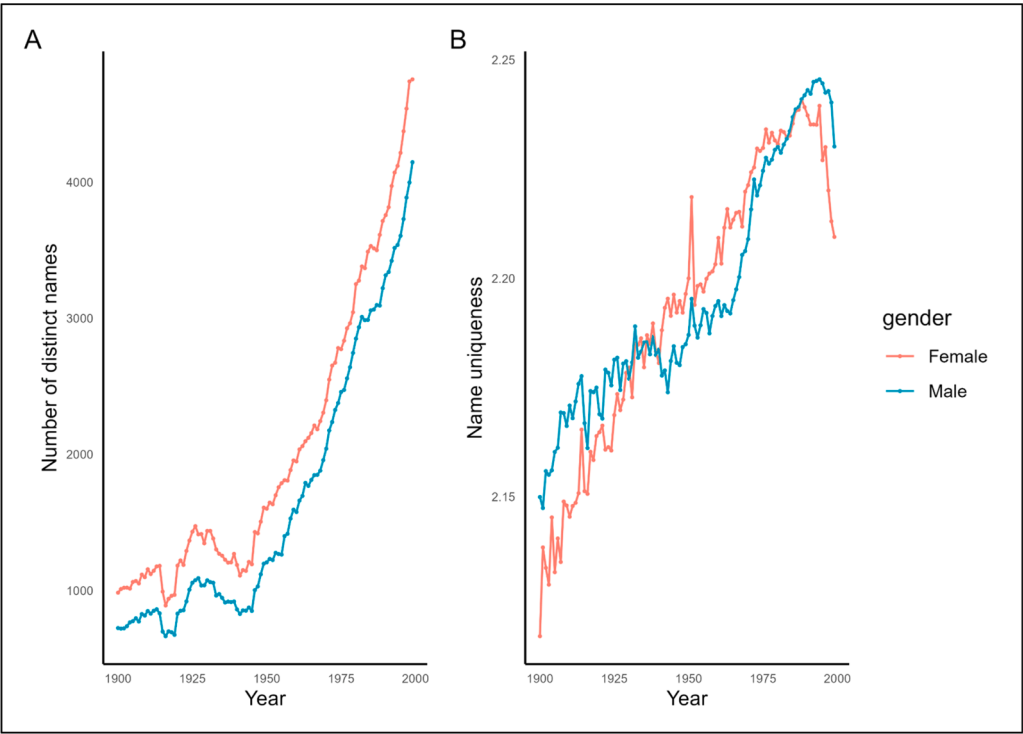


Fig. 3. Evolution of number of distinct names in the French national registry (A) versus the evolution of Name Uniqueness in the French notable people present in Wikidata (B).

2.3. Name uniqueness using Wikidata

Previous studies using national name registers have shown a general increase in the prevalence of unusual names. These studies typically use the traditional name metric like percentage of the top 5, 10, or 15 most common names in a given society, or number of distinctive names as an index of individualism (Mignot, 2022; Varnum and Kitayama, 2011). In this validation study, our aim is to test whether the evolution of Name Uniqueness among notable individuals in Wikipedia mirrors the evolution of Name Uniqueness among the general population in national registers.

Using the full French population registry from 1900–2000 (Mignot, 2022), we analyzed the annual number of distinct first names and its correlation with Name Uniqueness metrics derived from Wikidata. The results show a statistically significant positive correlation ($\beta = 0.5863$, $p < 0.001$) between registry-based name diversity (Fig. 3A) and crowd-sourced Name Uniqueness measures (Fig. 3B), supporting our prediction that Name Uniqueness based on an elite subset can be generalized to the entire population.

3. Result

3.1. Main model

We calculated uniqueness scores for 21 countries around the globe, spanning 500 years and organised by decade. We observed an overall increase in Name Uniqueness over this period (see Fig. 4). For countries

with the most data, we could examine longer time periods and observe that this increase is not monotonous. Between 1500 and 1750, Name Uniqueness increased weakly in France and Spain, remained stable in the UK and The Netherlands, and decreased in Italy, Germany and Sweden. Interestingly, following the emergence of Protestantism in the 16th century, there was no increase in Name Uniqueness in Protestant countries such as Germany, Sweden, Denmark, The Netherlands and the United States. Name Uniqueness only increased later, during the 19th and 20th centuries, in parallel with the rest of the Western world.

We tested the modernisation theory, which states that the increase in Name Uniqueness and individualism is driven by economic development. To do so, we constructed a mixed-effects model in which GDP per capita was the main predictor, year was the control variable used to account for overall temporal trends and country was the random effect used to capture variation between countries.

$$U \sim GDP_{percapita} + Year + (1|Country)$$

We find that Name Uniqueness increases with economic growth ($\beta = 0.26$, $p < 0.001$, $R^2 = 0.75$, see Table S1). This supports our prediction that Name Uniqueness increases alongside the economy.

3.2. Robustness check

To validate our findings, we conducted several robustness checks to ensure the reliability of our results.

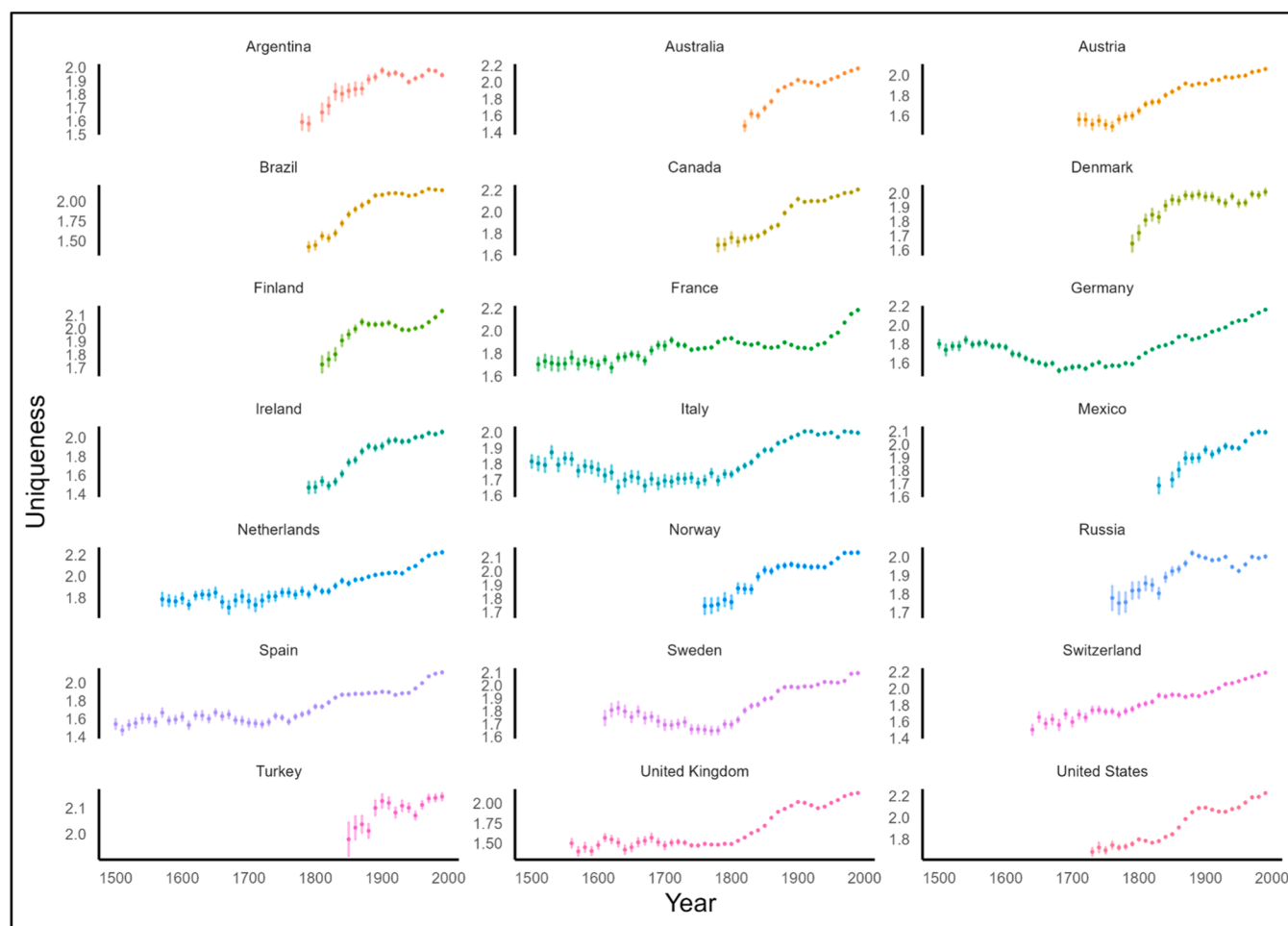


Fig. 4. Evolution of Name Uniqueness in the Western hemisphere. Each graph represents the evolution of one country, the bar is the confidence interval with 95 % confidence level, all countries show a general upward trend in Name Uniqueness.

3.2.1. Mixed-effects model (random slopes)

In addition to the base model we estimated a mixed-effects model with random slopes for GDP per capita and year, grouped by country. The inclusion of random slopes for GDP per capita and year enabled us to assess how the relationship between these predictors and Name Uniqueness varied across countries. This analysis confirms the result of our main model ($\beta = 0.18$, $p < 0.001$, $R^2 = 0.90$, see Table S2).

3.2.2. Fixed effect model

We re-estimated the models using country as a fixed effect instead of a random effect, and the results remained robust ($\beta = 0.27$, $p < 0.001$, $R^2 = 0.77$, see Table S3).

3.2.3. Control for gender and occupation

Using another version of Wikidata, which includes verified biographical information on individuals (Laouenan et al., 2022), we extended our analysis to examine Name Uniqueness based on gender ($\beta = 0.23$, $p < 0.001$, $R^2 = 0.66$; see Table S4 and Fig. S2) and occupation ($\beta = 0.26$, $p < 0.001$, $R^2 = 0.70$; see Table S5 and Fig. S3). Consistent with our primary findings, both Name Uniqueness measures—based on occupational groups and gender—yielded significant association with GDP per capita, confirming the result of our main model.

3.2.4. Balanced sample analysis

To address the limitations posed by the unbalanced dataset, we created a balanced sample by restricting the dataset to observations from 1850 onwards. During this period, reliable GDP records were consistently available for all countries in the database, with the exception of Russia. Consequently, Russia was excluded from this analysis. Using this balanced sample, we re-estimated the main random intercept model ($\beta = 0.24$, $p < 0.001$, $R^2 = 0.77$; see Table S6) and the random slope model ($\beta = 0.27$, $p < 0.001$, $R^2 = 0.90$; see Table S7). In both cases, we obtained significant results, which confirm our findings.

3.2.5. Control for population

To address potential confounding effects related to population structure, we have included Population as an additional control variable in our analyses. Furthermore, to account for clan development dynamics, we computed Surname Uniqueness using the same methodology as for first-Name Uniqueness and added it as an additional control. Even after incorporating these controls, the effect of GDP per capita remains statistically significant and robust ($\beta = 0.25$, $p < 0.001$, $R^2 = 0.76$; see Table S8).

3.2.6. Granger causality test

We performed a Granger causality test as a complementary analysis by building a Vector Autoregression (VAR) model, using the aggregated yearly data and selecting the optimal lag length based on BIC (lag = 2). The test revealed no significant Granger causality between GDP and Name Uniqueness ($F = 0.481$, $df = 2, 82$, $p = 0.620$). A country-level analysis using France, which has the most complete time series, yielded consistent results (lag = 1; $F = 0.100$, $df = 1, 82$, $p = 0.753$). These results indicate that GDP does not cause a change strictly after 1 or 2 decades.

4. Discussion

In this article, we quantify the evolution of Name Uniqueness, a proxy for individualism, over 500 years of global history. The Methods section demonstrates the reliability of using a sample of notable individuals from Wikidata to measure Name Uniqueness at the national level. Specifically, we successfully replicated results obtained from alternative name metrics that relied on full-sample name registers using the Name Uniqueness index. This consistency validates the use of Wikidata as a reliable source for studying Name Uniqueness. Our main study shows that individualism increased prior to the contemporary

period, aligning with qualitative assessments made by historians.

In line with modernisation theory (Inglehart, 2018), our findings also demonstrate that Name Uniqueness, a prevalent indicator of individualism, exhibits a positive correlation with GDP per capita between 1500 and 2000, reflecting contemporary trends. This result remains robust when controlling for Occupation and Gender, Surname Uniqueness, and Population. As societies modernise, the general level of affluence in the environment increases, enabling individuals to gain greater social mobility and prioritise personal goals, self-expression and unique aspirations over communal or familial expectations (Boon-Falleur et al., 2024; Inglehart, 2018). The rise of education, urbanisation and market economies also creates opportunities for people to make independent life choices, whether regarding careers, relationships or lifestyles, thereby driving the shift from collectivist to individualist values. Additionally, this research contributes to the long-standing social science debate about the role of Protestantism in fostering individualism. Contrary to Weber's hypothesis, our findings suggest that the adoption of Protestantism is associated with a decline in individualism rather than an increase. One plausible explanation is that the spread of Protestantism was followed by the Wars of Religion, which disproportionately affected Protestant countries—except for England, where Protestantism was imposed at a national level. This conflict resulted in significant population losses—up to 40 % of the German population—and a sharp decrease in GDP per capita (Koch et al., 2024; Pfister, 2022). According to modernisation theory, such setbacks may have led to a decline in individualism, as economic instability tends to reduce the emphasis on personal autonomy and self-expression.

An alternative explanation is that the decline in Name Uniqueness observed at the beginning of our analysis period (the 1500s) may be attributed to nation-building processes. During this time, smaller administrative units merged and the concept of the nation-state became more prominent. Although, it is worth noting that the decline is evident only in few countries, most notably Germany, Sweden and Italy, while other major European powers, such as France, Spain and England, continued to experience an increase in name diversity. Another issue with such structural explanations is the variation in nation-building processes across these regions. Sweden functioned as a cohesive nation-state from the beginning of the analysis period, whereas Germany and Italy did not become unified nation-states until the 19th century. Overall, economic development appears to be the most coherent and straightforward explanation for the long-term evolution of individualism.

The methodology of Name Uniqueness remains a key area for further investigation. Although the Name Uniqueness index has shown strong psychometric validity and theoretical relevance to individualism, its mathematical properties—particularly concerning sample size effects—are still not well understood. Subsequent studies should aim to formalize its statistical behavior and establish clearer guidelines for interpreting the measure in small populations.

Future research should also expand this investigation to non-Western countries, particularly in East Asia, where rich datasets are available. This region is particularly promising because cultural historians of China have suggested an early rise in individualism, which could be systematically tested using methods like Name Uniqueness (Brindley, 2010; Hegel and Hessney, 1985; Hinsch, 2015; Huang, 1995; Wang, 1994). Such studies could help determine whether the trends observed in the West are mirrored or distinct in other cultural contexts. Earlier periods presents an exciting opportunity to study the historical evolution of individualism. For example, Greek and Roman societies left behind a wealth of name inscriptions, which could offer valuable insights into how personal identity and autonomy were expressed in antiquity (Parker, 2019).

By analyzing patterns of Name Uniqueness and variability in these societies, researchers could explore whether individualistic tendencies began to emerge before the modern era and whether these patterns align with other cultural or economic developments of the time. These

extensions would not only validate the robustness of the method but also contribute to a broader understanding of the cultural evolution of individualism across diverse temporal and geographic contexts.

Declaration of competing interest

The authors declare no competing interest.

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Ethics and informed consent statement

This study utilized publicly available data from Wikidata. The dataset adhere to ethical guidelines and are openly accessible to researchers and the public. No direct interaction with participants was conducted, and no identifiable personal data was used. Therefore, additional ethics approval or informed consent procedures were not required.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.cresp.2025.100235](https://doi.org/10.1016/j.cresp.2025.100235).

Data availability

All data, materials, and code used in this study are publicly available and have been deposited in the Open Science Framework (OSF) repository: Project Name [Dataset]. OSF <https://osf.io/rzga2/>.

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